E-WASTE COMPLIANCE GAPS AND TRACEABILITY OPPORTUNITIES

Sahasra • Centum • Avalon • Tata Electronics • Dixon

Company-wise assessment based on publicly disclosed information, with a proposed digital compliance and traceability model

ABSTRACT

This study examines publicly visible e-waste compliance practices and information gaps associated with five electronics manufacturing companies: Sahasra, Centum, Avalon Technologies, Tata Electronics and Dixon Technologies. The purpose is not to allege legal violations, but to identify areas where compliance, end-of-life handling, EPR evidence and waste traceability are not sufficiently demonstrated through public disclosures. The analysis also translates these gaps into a practical digital-platform opportunity covering collection, logistics, authorised recycling, EPR evidence and audit trails.

Keywords: E-waste, EPR, compliance, authorised recycler, traceability, recycling, manufacturer, audit trail.

1. INTRODUCTION

Electronic manufacturing creates both production-related waste and, over the product life cycle, end-of-life electronic waste. The supplied reference paper highlights that e-waste management requires effective collection, authorised recycling, monitoring and stronger enforcement. It also stresses that informal handling can create environmental and health risks. filecite turn0file0 L648-L676

For a manufacturer or producer, the practical challenge is therefore not only whether a recycler exists, but whether the movement of each relevant waste stream can be demonstrated, documented and connected to applicable compliance obligations. The reference paper specifically identifies the need for stronger monitoring and a global/life-cycle tracking approach. filecite turn0file0 L780-L789

2. IMPORTANT INTERPRETATION

A compliance gap is not automatically a legal violation. The company-wise section below uses terms such as “not publicly demonstrated”, “requires verification” and “potential gap”. A confirmed violation should only be stated when supported by an official finding or enforcement action. Also, EPR applicability depends on the company's legal role and the particular products/EEE involved.

3. COMPANY-WISE COMPLIANCE GAPS

Sahasra — CPCB EPR / registration

Public indication: Earlier company disclosures indicated that Sahasra had applied for the relevant e-waste management system/registration.

Gap / verification point: The current registration status should be verified directly on the CPCB EPR portal; the historical disclosure should not be treated as current status.

Sahasra — Collection and recycling tracking

Public indication: The company describes e-waste collection/recycling through a recycling partner and reverse-logistics arrangements.

Gap / verification point: Detailed product-wise quantities collected, recycled and linked EPR certificates are not clearly demonstrated in the public information reviewed.

Centum — EPR applicability

Public indication: Centum's FY2024-25 reporting states EPR is “Not Applicable” to its activities and describes disposal through an approved recycler.

Gap / verification point: The legal classification of each activity/product should be verified before concluding that EPR is or is not applicable.

Centum — End-of-life take-back

Public indication: The company reports collection, segregation, storage and disposal, including sending e-waste to an approved recycler.

Gap / verification point: A consumer-facing/product take-back mechanism is not clearly demonstrated in the public disclosures reviewed.

Avalon Technologies — EPR / registration

Public indication: An earlier annual report indicated that EPR was applicable and registration was in process.

Gap / verification point: The current registration/portal status needs verification; the historical statement alone cannot establish present non-compliance.

Avalon Technologies — E-waste disposal and EPR evidence

Public indication: The company reports selling e-waste to authorised recyclers.

Gap / verification point: Public disclosures do not clearly provide detailed EPR certificate, target-fulfilment and end-to-end traceability data.

Tata Electronics — EPR compliance

Public indication: Public material indicates environmental-compliance responsibilities covering e-waste/EPR-related activities.

Gap / verification point: The public information reviewed does not provide enough detail to independently verify every EPR target, certificate and waste movement.

Dixon Technologies — End-of-life products

Public indication: Dixon has disclosed that it does not reclaim end-of-life products in its B2B model.

Gap / verification point: Direct take-back is a potential gap, but legal responsibility may sit with the brand/customer that is the Producer under the applicable rules.

Dixon Technologies — Recycled/reused material

Public indication: Sustainability reporting has disclosed periods/categories with nil recycled/reused input material.

Gap / verification point: This is a sustainability opportunity, not automatically an E-Waste Rules violation.

4. COMMON SYSTEM-LEVEL GAPS

Across the five cases, the most useful project opportunities are not accusations of non-compliance but transparency and traceability problems. The supplied reference paper similarly argues for stronger monitoring, verifiable disposal procedures, life-cycle tracking and better collection/recycling practices. filecite turn0file0 L780-L789

- EPR tracking: Central dashboard for obligations, targets, certificates and completion status.
- End-to-end traceability: Track a waste batch from source/collection through transport to authorised recycler.
- Recycler verification: Maintain verified recycler records and validate handover to authorised facilities.
- EPR certificate visibility: Link recycling transactions to supporting EPR certificate evidence.
- Product-level data: Capture product/category, quantity, source, age and end-of-life information.
- Manufacturing scrap vs. end-of-life: Separate factory-generated production waste from returned/consumer end-of-life electronics.
- Recycler performance: Track received, processed, recovered and disposed quantities where data is available.
- Compliance dashboard: Show completed, pending and at-risk compliance actions.
- Automated reporting: Reduce manual preparation of compliance records and periodic reports.

- Final-destination proof: Maintain a digital audit trail showing the final destination and processing evidence.

5. PROPOSED DIGITAL MVP

A practical MVP can act as a shared compliance and traceability layer rather than replacing the existing regulatory system. Each e-waste batch receives a unique digital identity and moves through defined stages.

1. Manufacturer/Producer creates the relevant product and compliance record.
2. E-waste batch is created with category, quantity, source and date.
3. Collection point/logistics partner scans or updates the batch.
4. Transport movement and supporting documents are recorded.
5. Authorised recycler receives and processes the batch.
6. Recycler records processing/recovery information and certificate details.
7. Platform matches recycling activity with applicable EPR obligations.
8. Dashboard displays completed, pending and at-risk compliance items.
9. Audit trail preserves evidence for every stage.

6. WHY THIS IS A VALID PROJECT PROBLEM

The reference paper concludes that India has rules for proper collection and authorised recycling but that successful enforcement and stronger monitoring remain necessary. It also notes that only a limited portion of e-waste is recycled through officially approved centres and calls for improved collection, infrastructure and monitoring. filecite turn0file0 L646-L676

This creates a defensible project proposition: instead of claiming that manufacturers are failing to obey the law, build a system that makes compliance measurable, traceable, verifiable and easier to audit. Such a platform can integrate manufacturer records, collection, transportation, recycler processing and EPR evidence into one digital trail.

7. CLAIMS TO AVOID

- Do not state that any of the five companies is breaking the E-Waste Rules without an official finding.
- Do not assume every electronics manufacturer has identical EPR obligations.
- Do not treat missing public information as proof that the company does not perform the activity internally.
- Use “publicly not demonstrated”, “requires verification”, “transparency gap” or “potential improvement area”.

8. REGULATORY / RESEARCH CONTEXT

The supplied academic reference discusses India's evolving e-waste framework, including the earlier E-Waste (Management and Handling) Rules, 2011 and later developments such as the 2018 amendment requiring e-waste to be channelized toward authorised dismantlers and recyclers. filecite turn0file0 L595-L632 It also emphasizes EPR, life-cycle assessment, material-flow analysis and multi-criteria analysis as useful approaches for strengthening e-waste policy. filecite turn0file0 L667-L676

9. REFERENCES / VERIFICATION SOURCES

- CPCB E-Waste EPR Portal — https://eprewaste.cpcb.gov.in/
- CPCB E-Waste FAQ — https://eprewaste.cpcb.gov.in/assets/PDF/faqewaste.pdf
- Sahasra Electronics — E-Waste Management page
- Centum Electronics — Annual Report FY2024-25
- Avalon Technologies — Annual Report FY2024-25
- Tata Electronics — Caring for the Earth
- Dixon Technologies — Sustainability